

# Database Management System (DBMS) Architecture

Q. Explain the different types of Database Management System (DBMS) Architectures. Differentiate between Two-Tier and Three-Tier architecture.

## 📌 Definition to Remember

**DBMS Architecture** is the fundamental design structure of a database system that determines how data is stored, processed, and accessed by users and applications. It is classified into One-Tier, Two-Tier, and Three-Tier architectures based on how the application connects to the database.

## 1. Components of DBMS

A typical DBMS architecture consists of:

- \* **Users:** End-users, database administrators, and application programmers.
- \* **Database Application:** The software interacting with the database.
- \* **DBMS Software:** The engine that manages data storage and retrieval.
- \* **Physical Database:** The actual storage media (hard drives) where data resides.

## 2. Types of DBMS Architecture

DBMS architectures are classified into three tiers:

### 1. One-Tier Architecture

- The **user, application, and database** all reside on the **same machine**.
- Used for standalone desktop applications (e.g., MS Access, SQLite).
- **Advantage:** Simple to set up. **Disadvantage:** Not suitable for multiple users or networks.

### 2. Two-Tier Architecture (Client-Server)

- Divided into two parts: the **Client** (application interface) and the **Server** (DBMS + data).
- The client connects directly to the database server using APIs like **ODBC** or **JDBC**.
- Application logic runs on the **client machine** (called a "fat client").
- **Advantage:** Faster for small user counts. **Disadvantage:** Security risks; performance drops with many users.

### 3. Three-Tier Architecture

- Introduces a **middle layer (Application Server)** between client and database server.
- **Three Layers:**
  1. **Presentation Layer (Client Tier):** User interface (e.g., web browser) — no direct DB access.
  2. **Application Layer (Business Logic Tier):** Middle server that processes requests and enforces business rules.
  3. **Database Layer (Data Tier):** Backend server storing and managing data.
- **Advantage:** High security, highly scalable, best performance for large user bases.

```

[Client Tier]
Web Browser / App
    |
    ▼
[Application Layer]
Business Logic Server
    |
    ▼
[Database Layer]
DBMS + Physical DB

```

### 3. Difference: Two-Tier vs Three-Tier

| Feature                  | Two-Tier Architecture    | Three-Tier Architecture                       |
|--------------------------|--------------------------|-----------------------------------------------|
| <b>Layers</b>            | Client + Database Server | Client + Application Server + Database Server |
| <b>Application Logic</b> | On the Client machine    | On the middle Application Server              |
| <b>Security</b>          | Lower (direct DB access) | Higher (DB hidden behind app server)          |
| <b>Scalability</b>       | Poor                     | Excellent                                     |
| <b>Best Suited For</b>   | Small LAN applications   | Web/enterprise applications                   |

#### ★ Must-Write Points (for 10 marks)

1. DBMS Architecture defines how users, applications, and the database interact.
2. One-Tier: single machine; Two-Tier: client-server; Three-Tier: client + app server + DB server.
3. Two-Tier uses ODBC/JDBC for direct client-to-database connection — called a "fat client."
4. Three-Tier's middle Application Layer enforces business logic and acts as a security buffer.
5. Three-Tier is more secure because the client never directly accesses the database.
6. Two-Tier suits small networks; Three-Tier suits large-scale web applications.
7. Three-Tier is the modern industry standard for enterprise and web-based systems.

#### ⚡ Quick Recall

One-Tier (1 machine) → Two-Tier (Client + DB Server, ODBC/JDBC) → Three-Tier (Client + App Server + DB, scalable & secure)